Setting up volumes

Volumes are views of cloud object storage. We create project-specific volumes. Use them for path-based access to structured or unstructured data. Volumes sit under a database in a catalog and are used to manage and provide access to data files. You can govern access to volumes using `GRANT` statements. Volumes provide scalable file-based storage without sacrificing governance. Often, we use unstructured, semi-structured, or non-tabular data in machine learning. Images are a good example of unstructured, non-tabular data that we will use for the Multilabel Image Classification project. To work with these images, the Multilabel Image Classification project uses volumes.

Starting the projects

We have planned our platform and set up our workspace environment. Next, let's work through each project. In GitHub, you will see that each chapter has a folder containing folders corresponding to each project. When we refer to the notebooks by name, we assume you are in the appropriate chapter and project folder. For example, this chapter has the first notebook:

```
Chapter 2: Designing Databricks: Day One/
  Project: Favorita Store Sales - TimeSeries Forecasting/
CH2-01-Downloading_Sales_Forecast_Data
```

We refer to the notebook by only the filename itself, `CH2-01-Downloading_Sales_Forecast_Data`. Let's jump into the first project.

Project: Favorita Store Sales – Time Series Forecasting

Recall from *Chapter 1* that we use a Kaggle-provided dataset to forecast sales. In this chapter, we download our data from the Kaggle website. To follow along in your workspace, please open the following notebook:

- `CH2-01-Downloading Sales Forecast Data`

In the notebook, and as well as the code here in *Figures 2.18 and 2.19*, we set our path and download our data from Kaggle.

First, we designate `raw_data_path` to store the files.

```
raw_data_path = volume_file_path + 'raw_data'

dbutils.fs.mkdirs(raw_data_path)
```

Figure 2.18 – Setting the path for our volume